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BE (7th Semester)

Examination, Nov.-Dec., 2013

Branch : Electrical, EEE

POWER APPARATUS SYSTEM

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part (a) from each question is compulsory.

(ii) Attempt any two parts from b, c and d in each question.

(iii) Make suitable assumptions if required.

Q. 1. (a) Pin type insulators are not used above 50 KV, why ? 2

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(b) Why the insulators are made of porcelain ? Explain methods of improving string efficiency ? 7

(c) Find the voltage distribution and string efficiency of a 03 unit suspension if capacitance of the link-pin to earth and to the line, are 20% and 10% respectively of the self capacitance of each unit.

If a guard ring increases capacitance to line of lower link pin to 35% of self capacitance of each unit, find the redistribution of voltage and string efficiency. 7

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(d) A string of 07 suspension insulators is to be fitted with a guard ring. If, pin to earth capacitance is = C, find values of line to pin capacitance that would give a uniform voltage distribution over the string. 7

Q. 2. (a) Define "substation", on what basis it is classified ? 2

(b) Write the functions as well as location of various substation equipments in a tabular form. 7

(c) Explain single bus-bar scheme. Also write its merits and demerits. 7

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(d) Explain the design of an earthing for a substation. 7

Q. 3. (a) Why earthing is done ? What are its types ? 2

(b) Explain the grounding through an earthing transformer. 7

(c) In a 50 Hz overhead line, capacitance of one line to earth was 1.5 μF. It was decided to use an earth-fault neutralizer. Calculate the reactance to neutralise the capacitance of : 7

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(i) 100% of length of line,

(ii) 90% of length of line,

(iii) 80% of length of line.

(d) Explain the 'Peterson-Coil" in detail. 7

Q. 4. (a) Write the principle of non-shielding method. 2

(b) A thyrite lightning arrestor has the characteristics :

$RI^{0.72}$ = 72,000, what is the ratio of the voltages appearing at the end of line having

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surge impedance of 500 Ω due to a 500 kV surge, when : 7

(i) the line is open circuited

(ii) the line is terminated by arrestor

(c) What are surge-diverters ? Explain in detail with the help of suitable diagrams.

(d) Explain the mechanism of lightning discharge.

Q. 5. (a) Define outage.

(b) Explain the "2-state model" in detail.

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(c) Explain the approximate method for calculation of reliability. 7

(d) How does a reliability is planned for the prepration of reliability models ? 7

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